

STAT6061/STAT5008 – Causal Inference

Part 4-3. Instrumental Variable

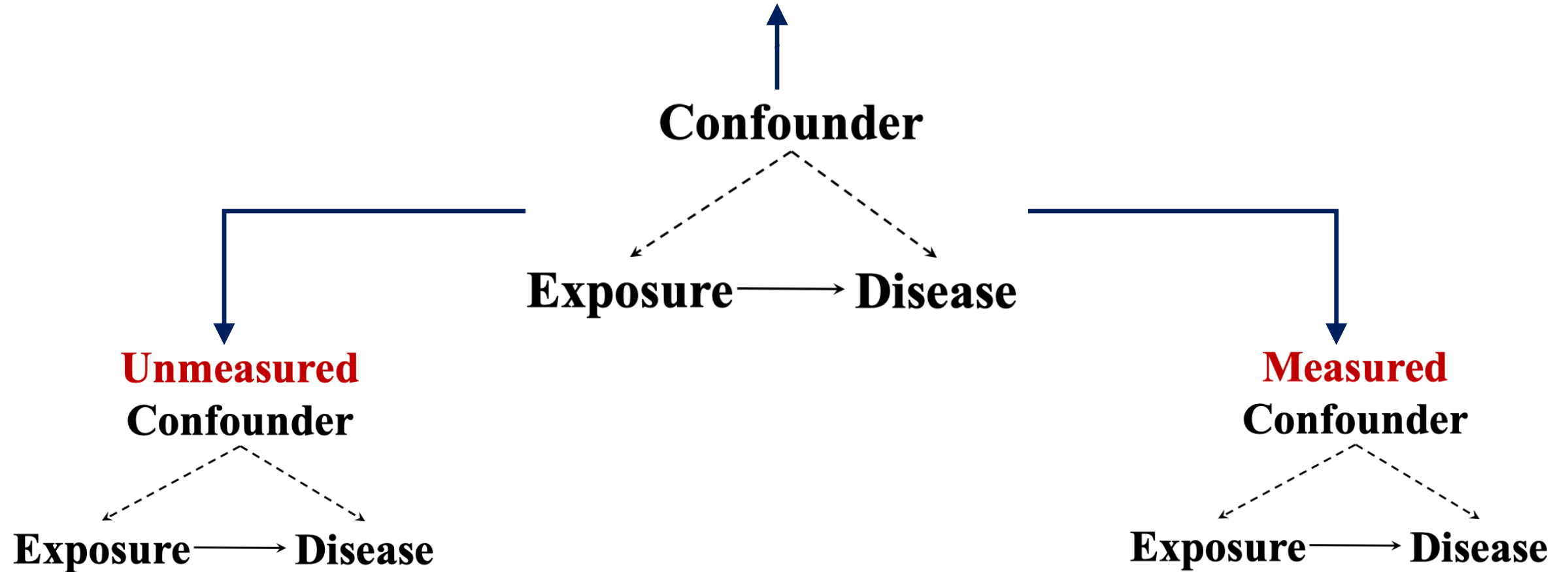
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- *Randomized Experiments*

Exposure —→ **Disease**



- *Front-door criterion*
- *DiD method*
- *Instrumental Variable (Mendelian Randomization)*

- *Standardization (g-formula)*
- *IPW*
- *Matching*

Instrumental variable

(Angrist, Imbens, and Rubin ,1996)

Definition 4.1 (Instrumental Variable).

Let A denote the treatment (or exposure), Y the outcome, and Z an observed variable. We say that Z is a valid *instrumental variable* for estimating the causal effect of A on Y if the following **core IV assumptions** hold:

1. **Relevance:** $Cov(Z, A) \neq 0$

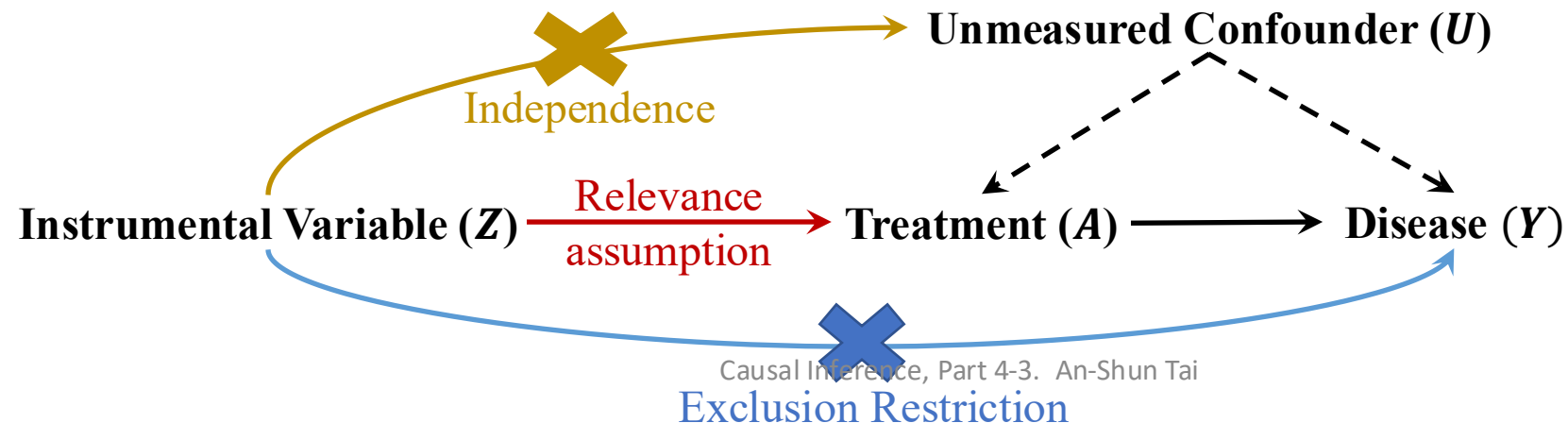
i.e., the instrument Z is associated with the treatment A .

2. **Exclusion Restriction (ER):** $Y(a, z) = Y(a, z') = Y(a)$ for all z and z'

i.e., the instrument Z affects the outcome Y only through the treatment A .

3. **Independence (or Unconfoundedness):** $Z \perp U$,

i.e., the instrument Z is independent of unmeasured confounders U that affect both A and Y .



Intuitive explanation of the three core IV assumptions

- Under the three core assumptions, we can think of Z as a **natural experiment**:
 - It induces exogenous variation in A that is independent of confounding and does not affect Y through other channels.
 - The observed effect of Z on Y reflects the portion of the effect of A on Y that is not contaminated by confounding.
- Example: High-Level neonatal intensive care unit (NICU) and in-hospital mortality (Lorch *et al.*, 2004)
 - Goal: Compare mortality for premature babies in high-level vs low-level NICUs
(High-level = mechanical ventilation & ≥ 50 premature infants/year)
 - Confounding effect: Sicker babies $\rightarrow$ more likely to be in high-level NICUs & more likely to die.
(Unmeasured confounders: fetal heart rate, severity of maternal conditions)
 - IV: Excess travel time ($Z=1$: extra travel time to high-level NICU ≤ 10 min; $Z=0$: extra travel time > 10 min)
Relevance: closer hospitals more likely chosen
Independence: assumes geography unrelated to unobserved severity
Exclusion Restriction: travel time does not directly impact mortality

Sources of instruments in health studies (Baiocchi, Cheng, and Small, 2014)

1. Randomized Encouragement

Example: Encouragement to quit smoking during pregnancy

Key idea: Randomly assigned encouragement (not the treatment itself) serves as the IV

2. Distance-Based Instruments

Example: Distance to nearest specialty hospital or high-level NICU

Key idea: Geographic proximity influences treatment choice but is assumed unrelated to unmeasured confounders

3. Preference-Based Instruments

Example: Physician, hospital, or regional variation in treatment styles

Key idea: Different providers have different tendencies (e.g., some prescribe more aggressively), which creates variation in treatment.

4. Time-Based Instruments

Example: Calendar time, policy implementation dates, or shifts in clinical practice

Key idea: Temporal changes affect treatment assignment but not potential outcomes directly

5. Genetic Instruments (Mendelian Randomization)

Example: Genetic variants (SNPs) influencing cholesterol levels or BMI

Key idea: Genes randomly assorted at conception serve as natural experiments for exposures

6. Administrative or Policy Variables

Example: Insurance plan features (e.g., copay levels, drug formularies)

Key idea: These affect access or adherence to treatment but are assumed not to directly affect the outcome

7. Admission Timing

Example: Time or day of hospital admission (weekday vs. weekend)

Key idea: Treatment protocols may vary by time, creating exogenous variation among otherwise similar patients

Partial identification of the ATE

(Swanson *et al.* 2018)

- The core IV assumptions define the conditions under which **an instrument is considered valid**, but they are not sufficient to identify the ATE from empirical data. Additional assumptions are required to **point-identify** the ATE. (Why? $(Z \rightarrow Y) - (Z \rightarrow A) \stackrel{?}{\Rightarrow} (A \rightarrow Y)$)
- Without additional assumptions, only partial identification of the ATE is possible, meaning the ATE can be bounded within an interval rather than point-identified.

Theorem 4.1 (Manski's worst-case bounds)

Suppose that the outcome and treatment are binary. Under minimal assumptions (e.g., no assumptions about the treatment selection mechanism), the ATE (τ) can be bounded by the observed data as follows:

$$L_{\tau} \leq \tau = \mathbb{E}(Y(1)) - \mathbb{E}(Y(0)) \leq U_{\tau}$$

where

$$U_{\tau} = \Pr(Y = 1|A = 1) \Pr(A = 1) + \Pr(A = 0) - \Pr(Y = 1|A = 0) \Pr(A = 0)$$

$$L_{\tau} = \Pr(Y = 1|A = 1) \Pr(A = 1) - [\Pr(Y = 1|A = 0) \Pr(A = 0) + \Pr(A = 0)]$$

- This bounds always have width 1 and thus include zero. $\Rightarrow$ It cannot determine the direction of the ATE.

Partial identification of the ATE (cont.)

(Swanson *et al.* 2018)

Theorem 4.2 (Robins-Manski bounds, natural bounds)

Suppose that the outcome, treatment, and IV are binary. Under core IV assumptions (e.g., relevance, exclusion restriction, independence), the ATE (τ) can be bounded by the observed data as follows:

$$L_{\tau}^{RM} \leq \tau = \mathbb{E}(Y(1)) - \mathbb{E}(Y(0)) \leq U_{\tau}^{RM}$$

where

$$U_{\tau}^{RM} = \max \left\{ \begin{array}{l} \Pr(Y = 1, A = 1|Z = 0) + \Pr(Y = 0, A = 0|Z = 0) \\ \Pr(Y = 1, A = 1|Z = 1) + \Pr(Y = 0, A = 0|Z = 1) \\ \Pr(Y = 1|Z = 0) - \Pr(Y = 1|Z = 1) + \Pr(Y = 0, A = 0|Z = 0) + \Pr(Y = 1, A = 1|Z = 1) \\ \Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0) + \Pr(Y = 0, A = 0|Z = 1) + \Pr(Y = 1, A = 1|Z = 0) \end{array} \right\}$$
$$L_{\tau}^{RM} = \max \left\{ \begin{array}{l} -\Pr(Y = 0, A = 1|Z = 0) - \Pr(Y = 1, A = 0|Z = 0) \\ -\Pr(Y = 0, A = 1|Z = 1) - \Pr(Y = 1, A = 0|Z = 1) \\ \Pr(Y = 1|Z = 0) - \Pr(Y = 1|Z = 1) - \Pr(Y = 1, A = 0|Z = 0) - \Pr(Y = 0, A = 1|Z = 1) \\ \Pr(Y = 1|Z = 1) - \Pr(Y = 1|Z = 0) - \Pr(Y = 1, A = 0|Z = 1) - \Pr(Y = 0, A = 1|Z = 0) \end{array} \right\}$$

- The width of the natural bounds is no greater than the sum of the noncompliance rates in each arm:

$$\Pr(A = 1|Z = 0) + \Pr(A = 0|Z = 1)$$

A fourth IV assumption

➤ There are two distinct types of fourth IV assumptions used for point identification:

1. Effect Homogeneity (Angrist, 1990):

Assumes that the treatment effect is constant (or mean-independent) across values of the instrument.

$$Y(1) - Y(0) \perp Z$$

Under this, the IV estimator identifies the **ATE**.

2. Monotonicity (Imbens & Angrist, 1994; Angrist, Imbens, & Rubin, 1996):

Assumes that the instrument shifts treatment in the same direction for all individuals (i.e., no defiers).

$$A(1) \geq A(0) \text{ for all units}$$

Under this, the IV identifies the **Local Average Treatment Effect (LATE)** for compliers.

A fourth IV assumption: Effect Homogeneity

- The original effect homogeneity assumption (i.e., $Y(1) - Y(0) \perp X$) is the most extreme and historically earliest form of homogeneity.

It is scientifically implausible for most treatments and outcomes and is impossible to satisfy for binary outcomes except under the sharp null or in cases of universal harm or benefit (**Why?**).

- Different versions of effect homogeneity

1. Additive homogeneity across levels of the instrument

$$\mathbb{E}(Y(1) - Y(0)|Z = 1, A = a) = \mathbb{E}(Y(1) - Y(0)|Z = 0, A = a) \text{ for } a = 0, 1$$

- A less extreme assumption than full constant effect.
- However, it is not intuitive, as experts may struggle to justify a constant causal effect across levels Z and A .

2. Effect homogeneity with respect to unmeasured confounders (U)

$$\mathbb{E}(Y(1) - Y(0)|U) = \mathbb{E}(Y(1) - Y(0))$$

- This is a more intuitive form, as it implies that U is not an effect modifier of the treatment effect.
- However, is typically untestable and often implausible.

A fourth IV assumption: Effect Homogeneity (cont.)

- The original effect homogeneity assumption (i.e., $Y(1) - Y(0) \perp Z$) is the most extreme and historically earliest form of homogeneity.

It is scientifically implausible for most treatments and outcomes and is impossible to satisfy for binary outcomes except under the sharp null or in cases of universal harm or benefit (Why?).

- Different versions of effect homogeneity

3. Homogeneity of the instrument–treatment association across levels of U

$$\mathbb{E}(A|Z = 1, U) - \mathbb{E}(A|Z = 0, U) = \mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)$$

- Testable implications: If some confounders are observed, the instrument–treatment difference should be the same across strata of those confounders.

➤ Remark

- Version 2 and 3 can be founded in Wang and Tchetgen Tchetgen (2018).
- Essentially, these three homogeneity assumptions hold under conditions of no interaction with the relevant variables.
 - Version 1: no $A - Z$ interaction on Y
 - Version 2: no $A - U$ interaction on Y
 - Version 3: no $Z - U$ interaction on A

Identification through effect homogeneity

Consider the following saturated, additive structural mean model for a dichotomous treatment A and an instrument Z

$$\mathbb{E}(Y(1) - Y(0)|A = 1, Z) = \beta_0 + \beta_1 Z$$

- β_0 represents the average treatment effect among treated individuals with $Z = 0$
- $\beta_0 + \beta_1$ is the effect among those with $Z = 1$.
- β_1 captures additive effect modification by Z .

➤ The saturated model can be rewritten as

$$\mathbb{E}(Y - Y(0)|A, Z) = A(\beta_0 + \beta_1 Z)$$

➤ The additive homogeneity across levels of the instrument (Version 1) implies $\beta_1 = 0$ and $\mathbb{E}(Y - Y(0)|A, Z) = A\beta_0$.
 $\Rightarrow \beta_0$ represents the ATT.

➤ Under the ER and independence assumptions, we have

$$\mathbb{E}(Y(0)|Z = 1) = \mathbb{E}(Y(0)|Z = 0).$$

It implies $\mathbb{E}(Y - A\beta_0|Z = 1) = \mathbb{E}(Y - A\beta_0|Z = 0)$.

➤ Solving the above equation, we obtain

$$\beta_0 = \frac{\mathbb{E}(Y|Z = 1) - \mathbb{E}(Y|Z = 0)}{\mathbb{E}(A|Z = 1) - \mathbb{E}(A|Z = 0)}$$

IV estimator

- Recall: if no unmeasured confounding assumption holds, the *ATE* can be estimated by

$$\widehat{\mathbb{E}}(Y|A = 1) - \widehat{\mathbb{E}}(Y|A = 0).$$

(Ignore the measured confounders C)

- Given a binary instrumental variable (Z), the **IV Estimator (Wald estimator)** is

$$\frac{\widehat{\mathbb{E}}(Y|Z = 1) - \widehat{\mathbb{E}}(Y|Z = 0)}{\widehat{\mathbb{E}}(A|Z = 1) - \widehat{\mathbb{E}}(A|Z = 0)}.$$

1. The IV Estimator is used to assess the *local average treatment effect* (LATE) rather than the *average treatment effect* (ATE). LATE is the treatment effect for the subset of the sample that takes the treatment if and only if they were assigned to the treatment.
2. The IV method can provide useful bounds on the average treatment effect for the whole population.

Ratio of Coefficients Method (Wald Method)

- The ratio of coefficients method (the Wald method) is the simplest way of estimating the causal effect of the exposure (A_i) on the outcome (Y_i), given a valid genetic variant (Z_i).

- **Continuous outcome and Dichotomous Z**

Ratio method estimate

$$\frac{\widehat{\mathbb{E}}(Y|Z = 1) - \widehat{\mathbb{E}}(Y|Z = 0)}{\widehat{\mathbb{E}}(A|Z = 1) - \widehat{\mathbb{E}}(A|Z = 0)}$$

Note: If no unmeasured confounding assumption holds, the ATE can be estimated by

$$\widehat{\mathbb{E}}(Y|A = 1) - \widehat{\mathbb{E}}(Y|A = 0).$$

(Ignore the measured confounders C)

Ratio of Coefficients Method (Wald Method)

- The ratio of coefficients method (the Wald method) is the simplest way of estimating the causal effect of the exposure (A_i) on the outcome (Y_i), given a valid genetic variant (Z_i).

- **Continuous outcome**

Suppose that

$$\begin{aligned} Y_i &= \alpha_y + \beta_{y|z} Z_i + \varepsilon_{1i} \\ A_i &= \alpha_a + \beta_{a|z} Z_i + \varepsilon_{2i} \end{aligned}$$

Ratio method estimate

$$\frac{\hat{\beta}_{y|z}}{\hat{\beta}_{a|z}} = \frac{Cov(Y, Z)}{Cov(A, Z)}$$

Ratio of Coefficients Method (Wald Method)

- The ratio of coefficients method (the Wald method) is the simplest way of estimating the causal effect of the exposure (A_i) on the outcome (Y_i), given a valid genetic variant (Z_i).

- **Binary outcome**

- Generally in epidemiological applications, disease is the outcome of interest. Disease outcomes are often dichotomous (Case: $Y_i = 1$; Control: $Y_i = 0$)

Ratio method log relative risk (RR) estimate =

$$\frac{\log \widehat{RR}_{y|z}}{\hat{\beta}_{a|z}}$$

Ratio method log odds ratio (OR) estimate =

$$\frac{\log \widehat{OR}_{y|z}}{\hat{\beta}_{a|z}}$$

Two Stage Least Square (TSLS or 2SLS)

- exposure (A_i); outcome (Y_i); valid genetic variant (Z_i).

TSLS is a computational method for calculating the MR causal estimator in a two-stage process.

Step 1: Regress A_i on Z_i , get fitted values $\hat{A}_i$.

$$A_i = \alpha_a + \beta_{a|z}Z_i + \varepsilon_{2i}$$

Step 2: Regress Y_i on $\hat{A}_i$.

$$Y_i = \delta_y + \theta\hat{A}_i + \varepsilon_{3i}$$

Wald Method vs. TSLS

Estimating the causal effect via Wald method

```
> data("iris")
> summary(iris)
  Sepal.Length   Sepal.Width   Petal.Length   Petal.Width   Species
Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa   :50
1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50
Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50
Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
> y.reg <- lm(Sepal.Length~Petal.Length,data=iris)
> x.reg <- lm(Sepal.Width~Petal.Length,data=iris)
> coef(y.reg)["Petal.Length"]/coef(x.reg)["Petal.Length"]
Petal.Length
-3.865587
```

Estimating the causal effect via TSLS

```
> x.reg <- lm(Sepal.Width~Petal.Length,data=iris)
> iris$Sepal.Width_prediction <- predict(x.reg)
> summary(lm(Sepal.Length~Sepal.Width_prediction,data=iris))

Call:
lm(formula = Sepal.Length ~ Sepal.Width_prediction, data = iris)

Residuals:
    Min       1Q   Median       3Q      Max
-1.24675 -0.29657 -0.01515  0.27676  1.00269

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    17.6617    0.5470   32.29  <2e-16 ***
Sepal.Width_prediction -3.8656    0.1786  -21.65  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

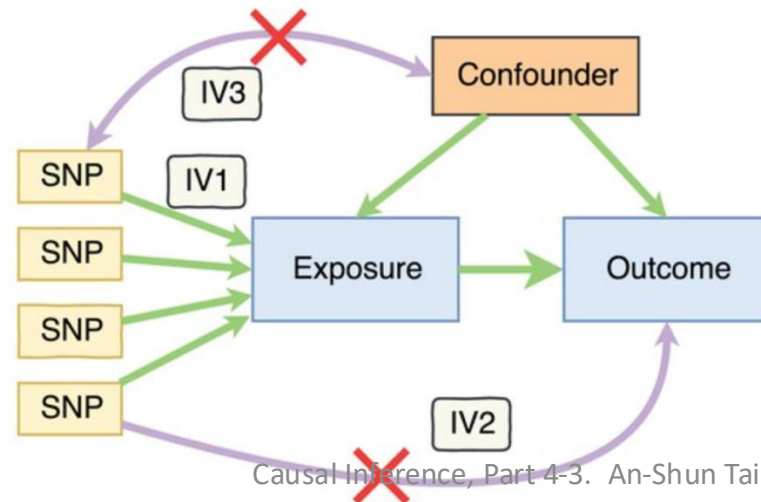
Residual standard error: 0.4071 on 148 degrees of freedom
Multiple R-squared:  0.76,    Adjusted R-squared:  0.7583
F-statistic: 468.6 on 1 and 148 DF,  p-value: < 2.2e-16
```

The estimates derived from both the Wald method and TSLS are identical!

Why do we need TSLS? and Why does TSLS work?

Why do we need TSLS ?

- TSLS does not inherently restrict the number of instrumental variables
- In many cases, a single genetic variant usually accounts for only a small fraction of the variation in a risk factor, often termed as "*weak instruments*," especially in small sample sizes.
- To address this limitation, researchers have devised approaches utilizing multiple genetic variants that collectively account for a greater portion of the variance in a risk factor compared to a single variant, thereby enhancing statistical power.



Hemani, Gibran, et al. "The MR-Base platform supports systematic causal inference across the human phenome." *elife* 7 (2018): e34408.

Why do we need TSLS? and Why does TSLS work?

Why does TSLS work?

- The slope coefficient obtained from regressing Y on A is
- $$\frac{Cov(Y, A)}{Var(A)}$$

Wald method

$$Y_i = \alpha_y + \beta_{y|z}Z_i + \varepsilon_{1i}$$

$$A_i = \alpha_a + \beta_{a|z}Z_i + \varepsilon_{2i}$$

Wald estimator

$$\frac{\hat{\beta}_{y|z}}{\hat{\beta}_{a|z}}$$

TSLS method

$$A_i = \alpha_a + \beta_{a|z}Z_i + \varepsilon_{2i}$$

Step 1: Regressing A_i on Z_i , we have

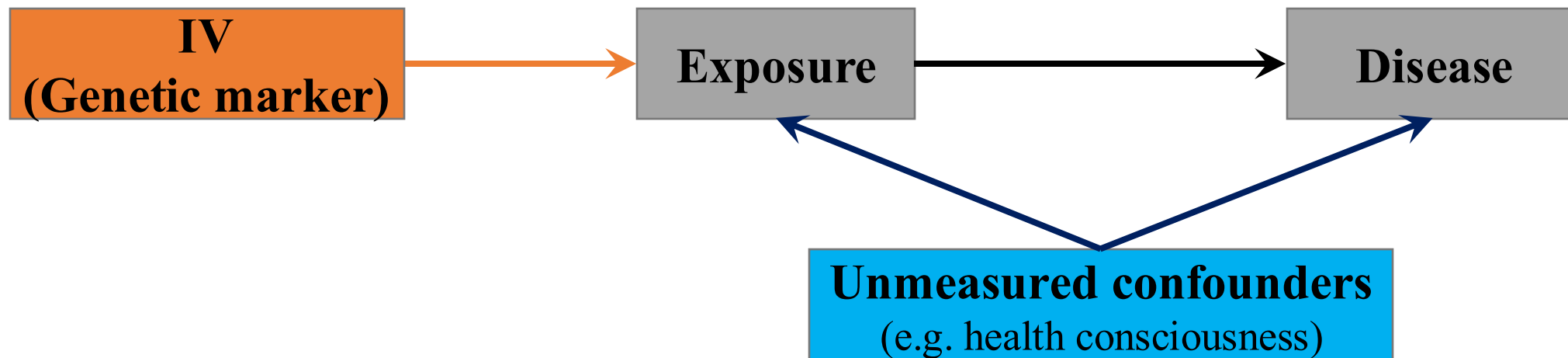
$$\hat{A}_i = \hat{\alpha}_a + \hat{\beta}_{a|z}Z_i$$

Step 2: Regressing Y_i on $\hat{A}_i$, slope coefficient is

$$\begin{aligned}\frac{Cov(Y, \hat{A})}{Var(\hat{A})} &= \frac{Cov(Y, \hat{\alpha}_a + \hat{\beta}_{a|z}Z)}{Var(\hat{\alpha}_a + \hat{\beta}_{a|z}Z)} = \frac{\hat{\beta}_{a|z}Cov(Y, Z)}{\hat{\beta}_{a|z}^2 Var(Z)} = \frac{\hat{\beta}_{a|z}\hat{\beta}_{y|z}Var(Z)}{\hat{\beta}_{a|z}^2 Var(Z)} \\ &= \frac{\hat{\beta}_{y|z}}{\hat{\beta}_{a|z}}\end{aligned}$$

Mendelian Randomization: Mendelian Randomization: A Tool for Causal Inference in Epidemiology

- Mendelian Randomization (MR) uses genetic variants as IVs to estimate the causal effect of an exposure (e.g., BMI) on an outcome (e.g., disease risk).
- Based on Mendel's laws, genetic variants are randomly assigned at conception, mimicking a randomized trial.



Relevance: genetic marker has a causal effect on exposure

Evaluation Focus

- In TSLS MR, focus lies on assessing the F-statistic and R^2 from the first-stage regression of the exposure on the genetic marker (SNP).

F-Statistic: Genetic IV Strength

- The F-statistic reflects the strength of the genetic marker, indicating the size and probability of bias in estimating causal association.
- A "weak" IV (genetic marker) is typically characterized by an F-statistic of less than 10 (rule of thumb).

R^2 : Determining Power

The R^2 from this regression determines statistical power to detect a causal effect, showing the proportion of variability in the exposure phenotype explained by the genetic variant.

Prior Knowledge

Good prior knowledge of variance explained by genetic markers helps determine IV strength and statistical power before study commencement.

Exclusion restriction (ER): genetic marker affects the outcome only through exposure.

Testing the exclusion restriction assumption

- The exclusion restriction assumption cannot be conclusively proven.
- However, statistical, theoretical or biological evidence may bolster ER if such evidence suggests direct pathways are unlikely.

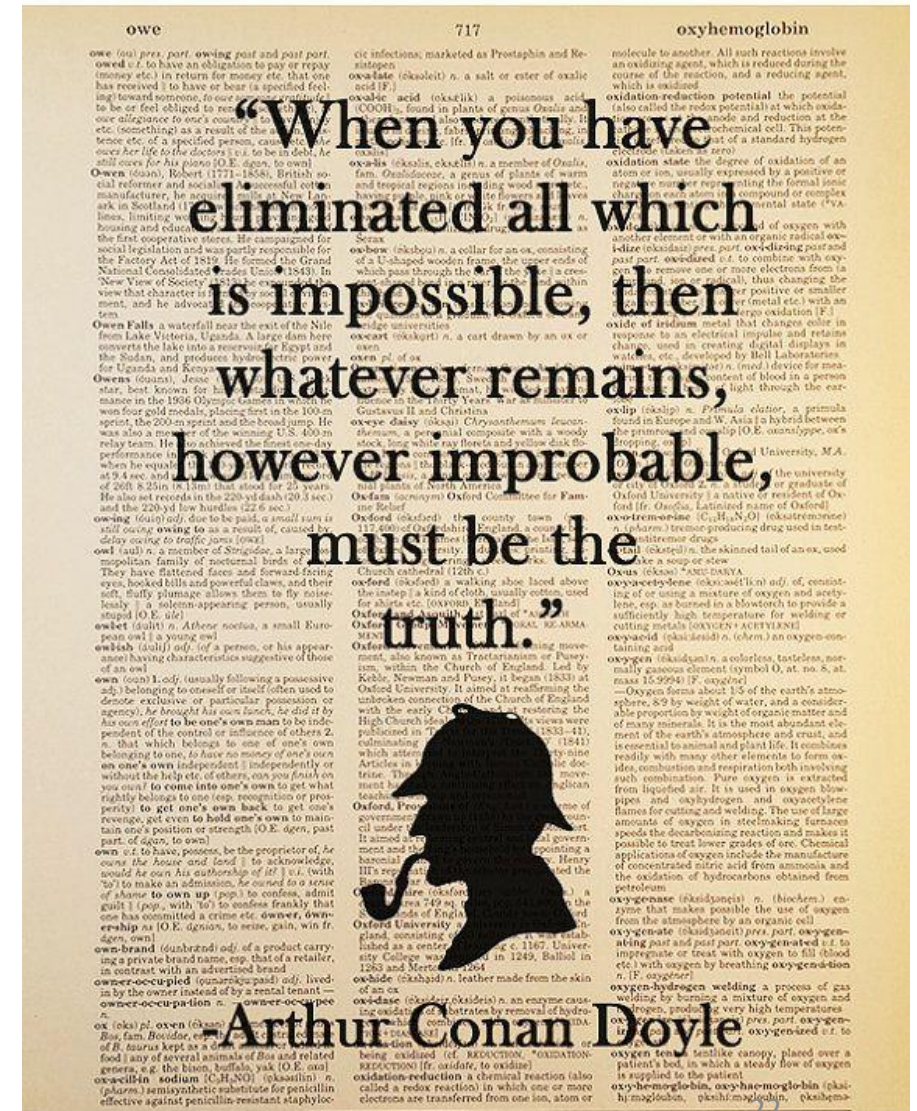
External evidence

External evidence can be used to quantify the genotype-phenotype association; to support the assumption of no other causal pathways between the genotype and the outcome.

Negative control outcomes and Negative control populations

Pleiotropy

- The phenomenon of a single genetic variant influencing multiple traits



Exchangeability assumption: genetic marker does not share common causes with the outcome.

- ✓ Genetic markers, inherited randomly at conception, are independent of environmental factors influencing the outcome.
- ✓ This randomness ensures genetic markers do not share common causes with the outcome, addressing confounding concerns present in observational studies.
- ✓ However, a key concern is **population stratification**, where ancestry may correlate with both the genetic marker and the outcome, potentially confounding MR results.
- ✓ Techniques such as ancestry adjustment or using ancestry-informative genetic markers can help mitigate population stratification effects in MR studies.

Negative control for ER and exchangeability

Negative control outcomes

- Negative control outcomes are likely to be affected by the same confounders as the outcome of interest but are unlikely to be directly affected by the exposure.
- If the genetic marker is associated with a negative control outcome, then this suggests that there may be residual confounding and that exchangeability assumption has been violated.

Negative control population

- A negative control population has a similar confounding structure as the population of interest but was not exposed to the treatment of interest.
- If the proposed genetic marker is associated with any outcomes in the negative control population, this suggests that it may be operating through another mechanism

Reference:

Davies, Neil M., et al. "How to compare instrumental variable and conventional regression analyses using negative controls and bias plots." *International Journal of Epidemiology* 46.6 (2017): 2067-2077.

One-sample MR

- Data from a single study: genetic variants, exposure, and outcome are measured in the same individuals.
- Allows MR and conventional epidemiological findings to be compared in the same individuals.
- Method for one-sample MR: TSLS method
- Individual level data vs. One-sample using summary data

Two-sample MR

- Data from two studies (from the same underlying population).
- Genetic variant (SNP)-exposure associations are estimated in one dataset, and genetic variant (SNP)-outcome associations are estimated in a second dataset.
- The two-sample MR analysis is usually performed using summary data.
 - Summary data are genetic association estimates from regression of the exposure or outcome on a genetic variant.
 - Two GWAS studies

Multiple instruments: Inverse variance weighted (IVW) method

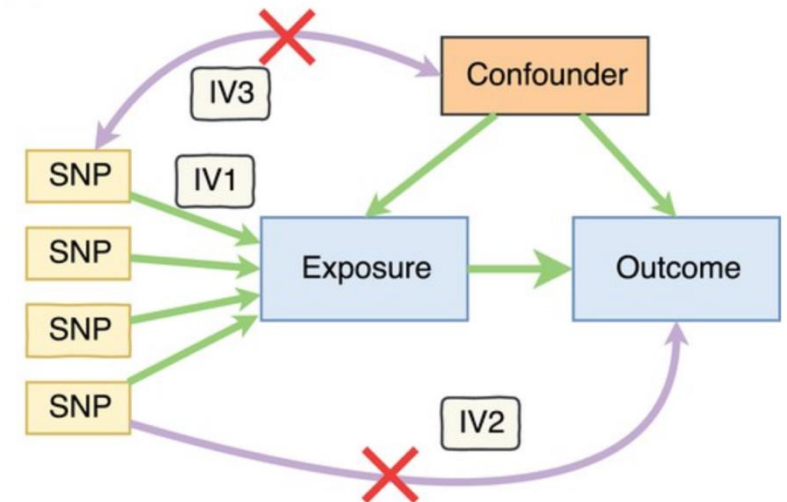
The traditional MR method can employ a meta-analysis approach along with the Wald method to estimate the causal effect derived from various genetic markers (SNPs).

$$Y_i = \alpha_{s,y} + \beta_{s,y|z}Z_{s,i} + \varepsilon_{1i}$$
$$A_i = \alpha_{s,a} + \beta_{s,a|z}Z_{s,i} + \varepsilon_{2i}$$

Wald estimator for SNP-s: $\hat{\theta}_s = \frac{\hat{\beta}_{s,y|z}}{\hat{\beta}_{s,a|z}} = \frac{Cov(Y, Z)}{Cov(A, Z)}$

$$\text{Wald estimator for pooled data (IVW): } \hat{\theta}_{pooled} = \frac{\sum_s (w_s \times \hat{\theta}_s)}{\sum_s (w_s)}$$

where $w_s = 1/Var(\hat{\theta}_s)$



Note: IVW estimates are analogous to performing a weighted linear regression of SNP-outcome associations against SNP-exposure associations, with the constraint that the intercept is zero.

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